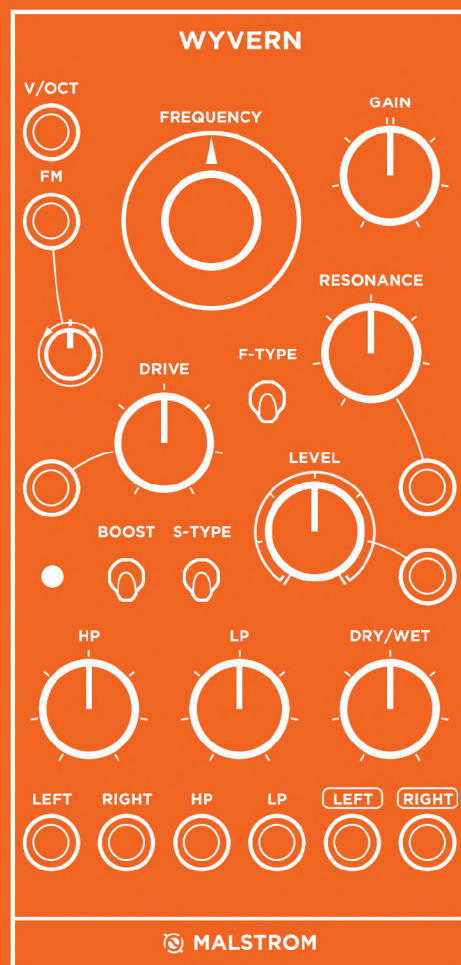


MALSTROM



WYVERN

USER MANUAL



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DESCRIPTION

Saturation is a sound designer's best friend. As applying saturation to a sound can make it sound warm, rich and defined. But without any other processing the opposite can happen; making the sound harsh, destroyed and undefined. An often used technique in music production is to saturate specific parts of the spectrum, adding harmonics to enrich the sound but not lose its original quality.

Wyvern takes this technique and makes it a creative sound design tool. When deconstructed Wyvern is built out of a specific chain: Filter EQ, Saturation, VCA, Dual Filter and Dry/Wet Balance. Different combinations of these stages can be used for a variety of sound design and patch situations. And it is all in stereo, so you can apply it to your final mix with ease too!

The Filter EQ stage has a high shelf, low shelf and bell curve to boost or diminish parts of the spectrum. When the gain is set to a minimum the EQ curves turn into lowpass, highpass and notch respectively. A Resonance control puts emphasis on a certain frequency and can make it scream.

This is then sent to a Saturation stage. The Drive control sets the amount of Saturation of one of the three Saturation types. Soft clipping puts emphasis on the third harmonic and compresses the sound. Hard clipping puts emphasis on the odd harmonics. Finally a special Allpass distortion adds phase shifts and alters transients. The Saturation stage also has a level control which can be used to decrease the gain after Saturation is applied and can function as a unipolar VCA. The Saturation stage is also accompanied with a clip indicator led and a boost switch to scale the input signal correctly or simply add more or less harmonics.

The Post-Saturation stage is a Highpass and Lowpass filter combo. Allowing you to select a frequency band from the previous stages. Along with a balance control to adjust the balance between the dry and wet signal.

Whether you are looking to destroy your sound or go for soft analog warmth, Wyvern has got you covered. Even when it is not used for saturation alone. It can be a tone controller, EQ, Filter, VCA and Phaser, all and anything in between. Making it a very fun and powerful sound design tool!

FEATURES

- Filter EQ with Low-shelf, High-shelf, Bell, Low-pass, High-pass and Notch curves.
- Resonance to add emphasis on a frequency or to make it scream.
- Soft clipping, Hard clipping and Allpass distortion for a variety of flavors.
- Boost switch and clip indicator led to set Saturation scale.
- Level and VCA to adjust level while Saturation remains the same.
- High-pass and Low-pass combo to select a part of spectrum after Saturation.
- Dry/Wet control to dial in the right amount of processed signal.
- Gain with increased and adjustable center detent.

SPECS

FORMAT
EURORACK

WIDTH
12HP

FRONT
20MM

BACK
28MM

WEIGHT
195GR

POWER @ +12V
165mA

POWER @ -12V
150mA

BOX CONTENT

1x Module
1x Ribbon power cable
4x Screws
4x Washers
2x Stickers

Note: Our packaging is made from recycled materials and is 100% recyclable. We also choose to provide digital manuals only!

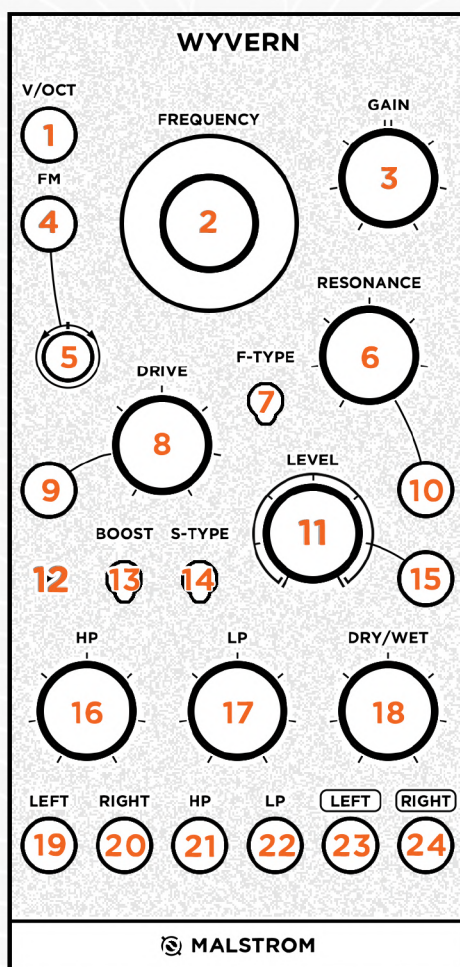
INSTALLATION

!!! WARNING !!! Be smart, turn power off before you start!

1. Check there is enough power available on the power supply.
2. Connect the 2x5 connector of the provided ribbon cable to the module.
3. Check if the red stripe on the ribbon cable is connected to the red stripe indicator on the module. The red stripe indicator is the -12V rail.
4. Connect the 2x8 connector of provided ribbon cable to the busboard.
5. Check if the red stripe on the ribbon cable is connected to the -12V position on the busboard.
6. Fasten the provided screws to lock the module onto the eurorack system rails.
7. Turn power on and enjoy!

Note: Malstrom modules have reverse polarity protection. Reversed power connection will not damage the module.

OVERVIEW



- | | |
|------------------------|--------------------------------|
| 1. V/Oct Input | 13. Boost Switch |
| 2. Frequency | 14. Saturation Type Switch |
| 3. Gain Knob | 15. Level CV Input |
| 4. FM CV Input | 16. Highpass Filter Knob |
| 5. FM Attenuverter | 17. Lowpass Filter Knob |
| 6. Resonance | 18. Dry/Wet Knob |
| 7. Filter Type Switch | 19. Left (or Mono) Audio Input |
| 8. Drive Knob | 20. Right Audio Input |
| 9. Drive CV Input | 21. Highpass CV Input |
| 10. Resonance CV Input | 22. Lowpass CV Input |
| 11. Level Knob | 23. Left Audio Output |
| 12. Clip Indicator LED | 24. Right Audio Output |

DETAILED VIEW

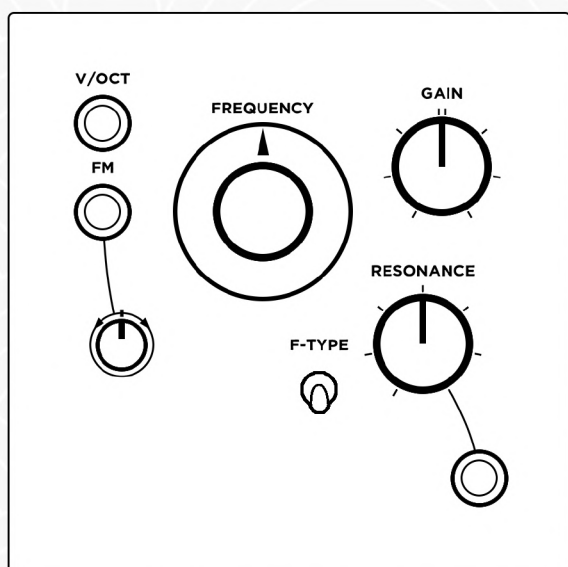
Wyvern can be divided into three main stages. The “Filter EQ” stage, the “Saturation” stage and the “Post-Saturation” stage.

The “Filter EQ” stage is used to change your input spectrum,

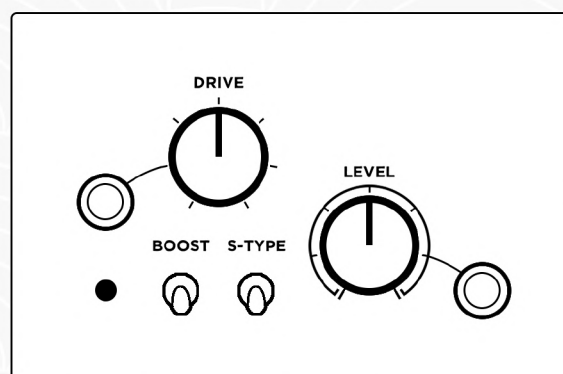
The “Saturation” stage is used to add harmonics and determine level.

The “Post-Saturation” stage is used to remove unwanted harmonics, and set the balance between your dry and wet signal.

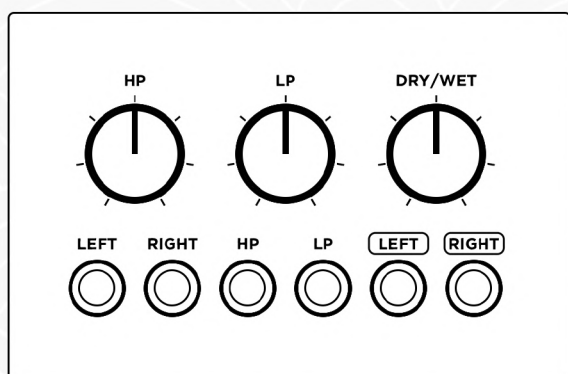
FILTER EQ



SATURATION



POST-SATURATION & IO

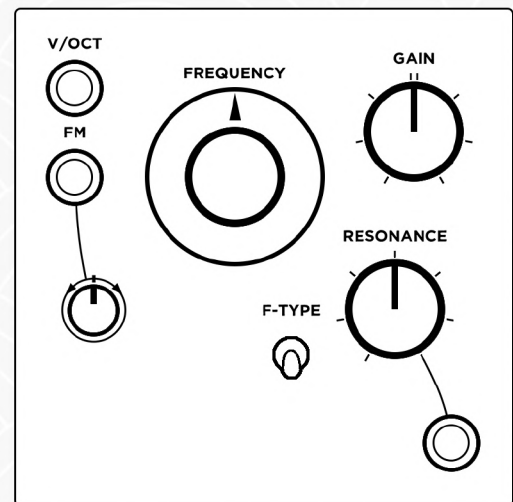


FILTER EQ

Let's follow the signal path and start at the top of the module. This part is a Filter EQ, meaning that it has functions of an EQ and a filter.

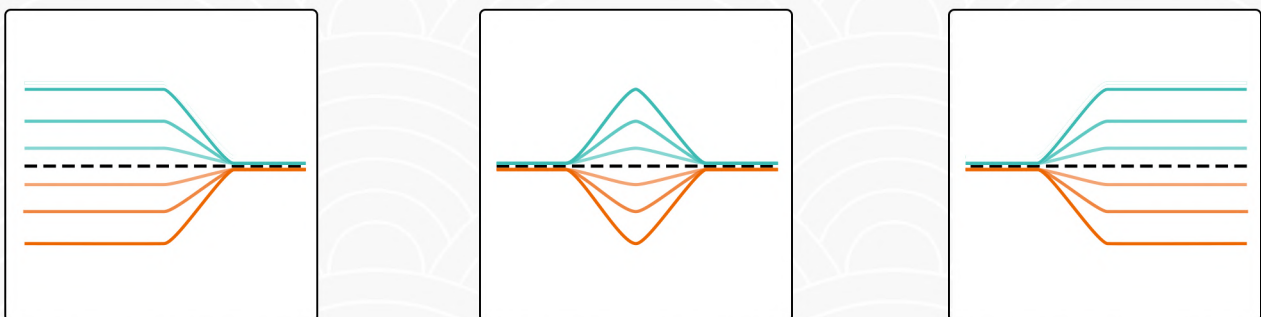
The graph on the right shows the controls in the Filter EQ stage. The Frequency Knob, FM Input and V/Oct Input are used to determine the filter's cutoff frequency. Resonance is the feedback amount of the filter and is capable of self-oscillation (see note 1).

The Filter Type Switch has three settings: High Shelf (switch to up position), Bell (switch to mid position) and Low Shelf (switch to down position).



The Gain Knob allows you to change the spectrum corresponding to the selected filter type and selected frequency. Move CW from the middle to boost frequencies up to +12dB, go CCW from the middle to decrease frequencies. The Gain Knob will also determine the gain of the Resonance.

When the gain is set to absolute minimum setting (fully CCW); new filter EQ shapes can be achieved due to phase cancellation. High Shelf turns into Lowpass, Bell turns into Notch (see note 2), and Low Shelf turns into Highpass. All of which are -6dB per octave.



Note 1: you can limit the amount of resonance, your audience may love you for it. See page 22 to learn more.

Note 2: In this low gain setting Resonance will always stay at the cutoff frequency, which is great of course! But this also means that high Resonance settings will cancel out the Notch filter.

SATURATION

From the Filter EQ your signal arrives at the Saturation stage. The correct internal scaling needs to be $\pm 10V$, the Boost Switch can help you set the correct gain before it is Saturated. However it can also be used to simply add harmonics via hard clipping, so have fun with it!

The Boost Switch has three settings: x1 gain (switch to down position), x2 gain (switch to mid position) and x4 gain (switch to up position).

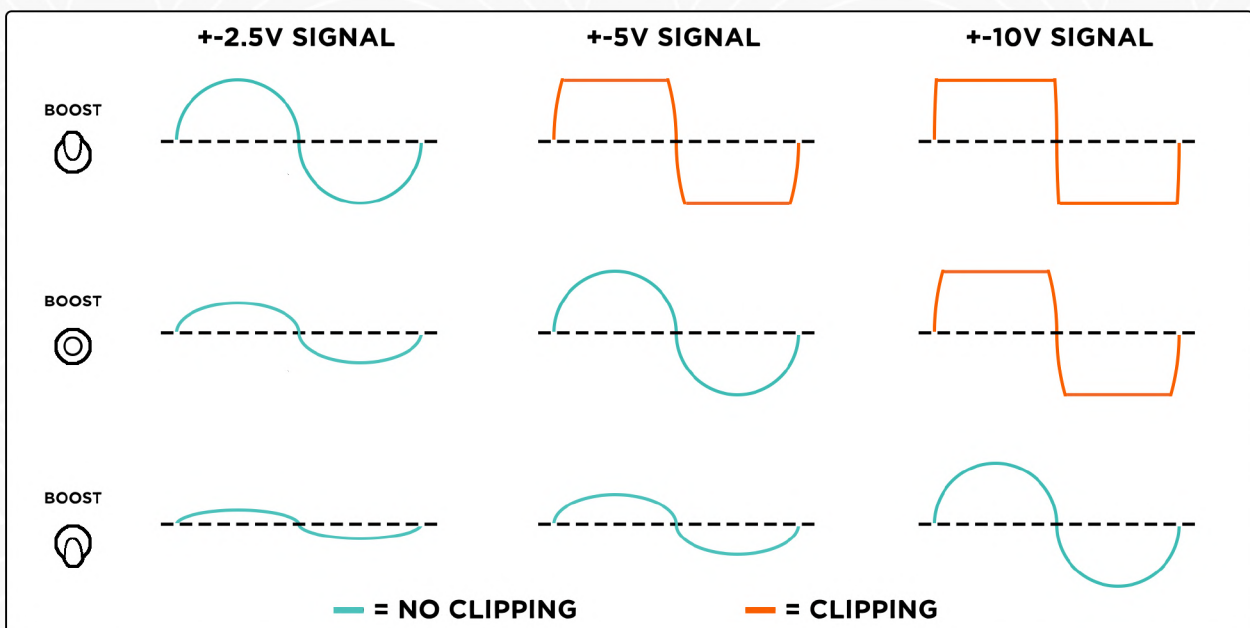
The x1 gain is for signals that range between $\pm 10V$. This setting will not boost and generally will not apply any distortion by itself (see note).

The x2 gain is for signals that range between $\pm 5V$. When this setting is used on a $\pm 10V$ signal; the signal is clipped and extra distortion is applied.

The x4 gain is for signals that range between $\pm 2.5V$. When this setting is used on a $\pm 10V$ or $\pm 5V$ signal; the signal is clipped and extra distortion is applied.

The Boost Switch is accompanied by a Clip Indicator LED, which shows you when there is clipping on either the left or right channel before Saturation.

The following graph shows when the Boost Switch is expected to clip.

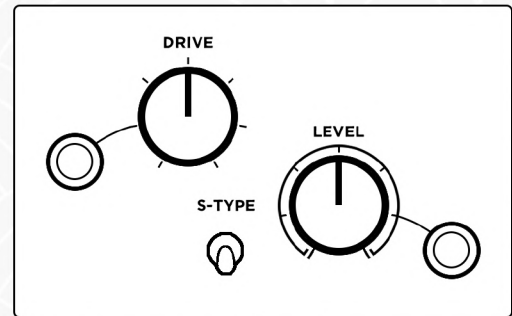


Note: The signal may have increased in gain at the Filter EQ stage. So keep your eye on the Clip Indicator LED!

The actual saturation is controlled via the Drive Knob and the Level Knob.

Drive will increase the input signal before it is sent to the selected saturation circuit.

Level can decrease the signal after the saturation process. Because saturation gives an increase in amplitude it may be useful to scale your signal down afterwards. When the Level Knob is fully CCW the signal will be silenced (see note 1).



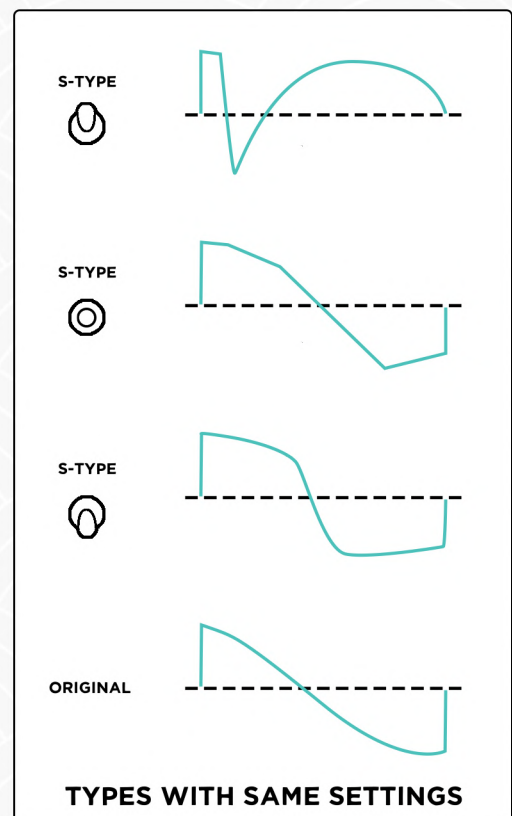
The Level CV Input functions as a VCA. When presented with a CV signal the Level Knob acts as a CV attenuator.

Wyvern gives you three flavors of Saturation: Allpass Distortion (Saturation Type Switch to up position), Hard Clipping (Saturation Type Switch to mid position) and Soft Clipping (Saturation Type Switch to down position),

Allpass distortion changes harmonic gain through a comb filter. It can also act as a waveshaper or phaser depending on how it is patched. The Allpass filter's timing corresponds to the Frequency Knob in the Filter EQ stage (see note 2).

Hard Clipping clips the wave and adds sharp edges, increasing even or odd harmonics depending on Drive setting.

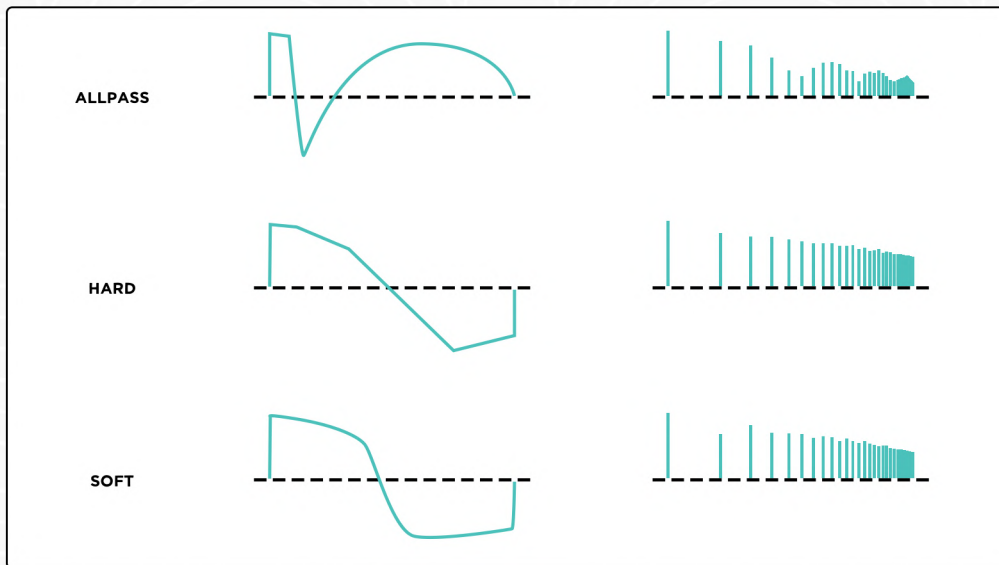
Soft Clipping is similar to Hard clipping but has rounded edges, by slewing the signal. It also does this asymmetrically. It has more emphasis on odd harmonics.



Note 1: Remember that this only applies to your wet signal!

Note 2: It is the only saturation type that responds to the Frequency knob.

Note 3: Even with the Drive Knob down, Soft Clipping will shape your sound.



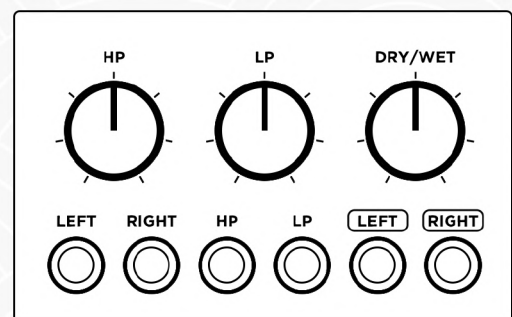
Here is how these waves would look in a spectrum analyzer. See how the Soft and Hard Clipping are similar, but clearly have a difference in the second and third harmonic. The Allpass Distortion entirely changes the gain structure of the harmonics known as a comb filter.

POST-SATURATION

The final stage is the Post-Saturation stage, this is where it all comes together.

It consists of a -6dB Highpass filter, a -6dB Lowpass filter and a Dry/Wet Balance control.

The Highpass and Lowpass filters can be used to select the frequency band you want to add to your dry signal. The knobs represent their cutoff frequencies, HP passes the entire signal when fully CCW and LP passes the entire signal when fully CW. It is good to note that these knobs have the same range, which makes it easier to see which part of the spectrum is selected (see note 1).

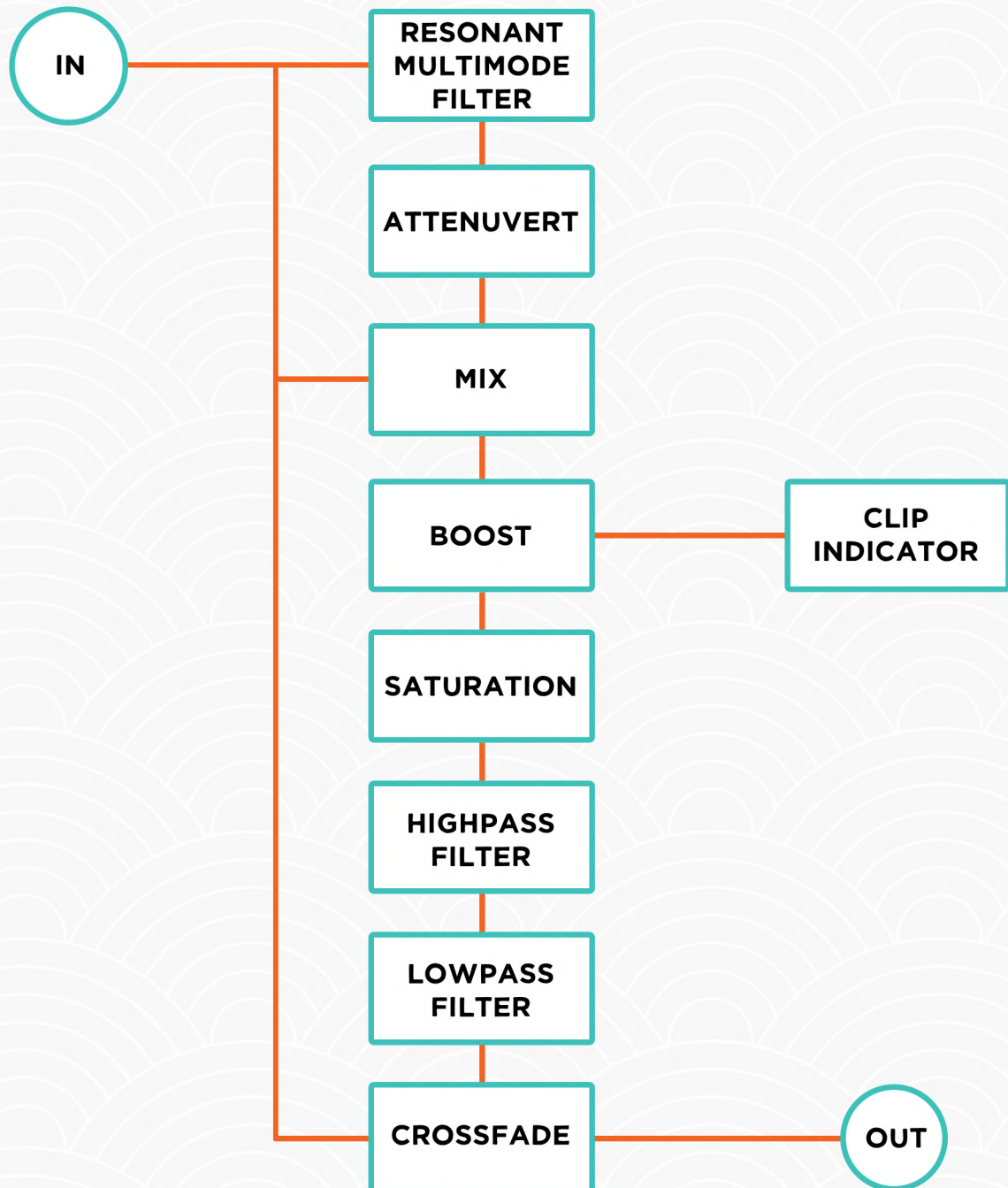


The Dry/Wet Balance Knob wraps it up, with your dry signal fully CCW and your wet signal fully CW.

Note 1: Remember that moving one knob passed the other will remove almost all frequencies. So a good rule is to keep HP left from LP, and LP right from HP!

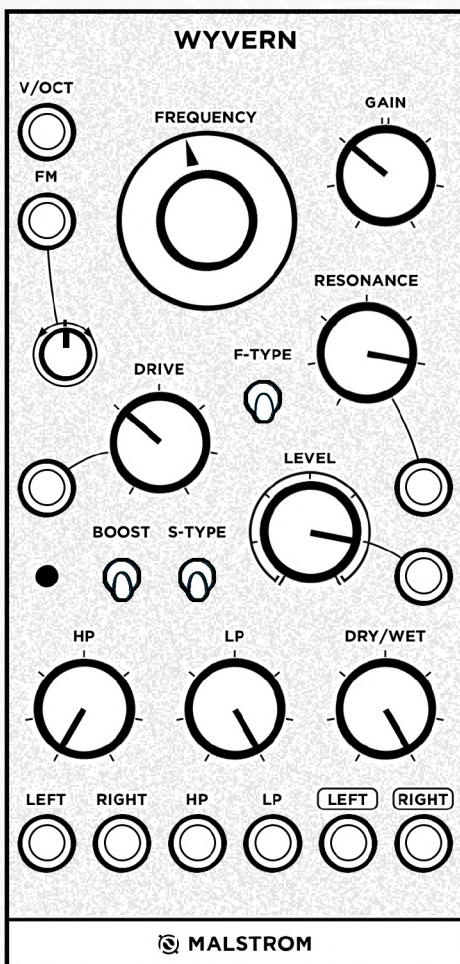
CIRCUIT DIAGRAM

The following diagram is a simplified version of how a single channel of Wyvern is connected. I.e. It is copied for the other channel.



REFERENCES

The following references are meant as starting off point to get saturation similar to famous saturation types. These references are not perfect in any way, and are not meant to be. As there are multiple ways to achieve these sounds. To get the wanted result you will need to adjust to taste and your input signal!



TAPE SATURATION

Tape Saturation can be described as having attenuation in the higher frequencies. So we use the Filter EQ in High Shelf and decrease the higher frequencies with the Gain Knob.

We will also add a bit of Resonance to increase gain around the cutoff frequency, which lies in the mid-range of the spectrum. Emulating tape speed artifacts.

Going to the Saturation Stage we will select the Soft Clipping setting, due to its slewing, emphasis on the third harmonic.

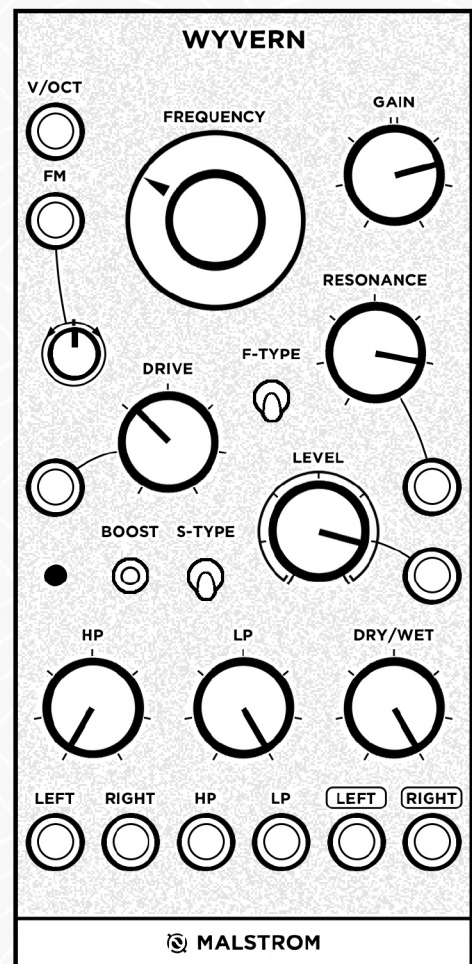
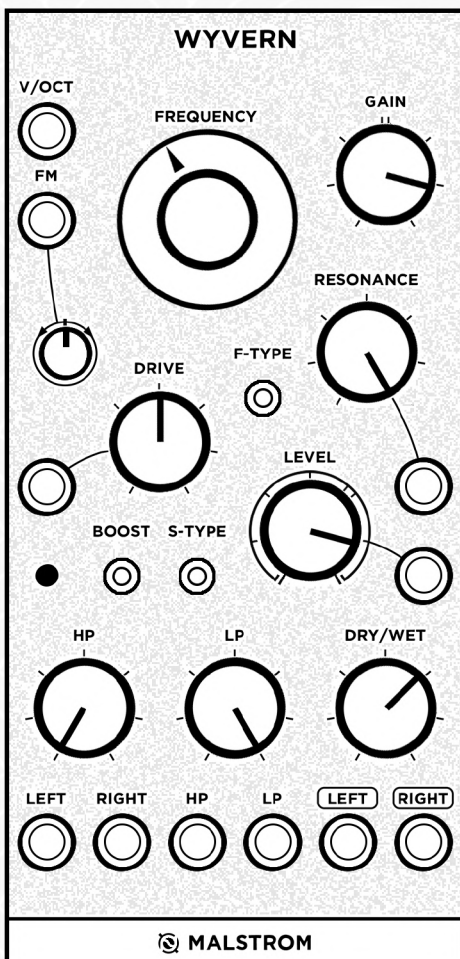
Tip: Alternatively we can use the Hard Clipping setting for its high frequency ringing!

TUBE SATURATION

Tube Saturation can be described as having amplification in the lower frequencies. So we use the Filter EQ in Low Shelf and increase the lower frequencies with the Gain Knob.

We will also add some Resonance to increase the gain around the cutoff frequency, in the low-range of the spectrum.

Going to the Saturation Stage we will again select the Soft-Clipping setting, because it slews and generates lower harmonics.



FUZZ DISTORTION

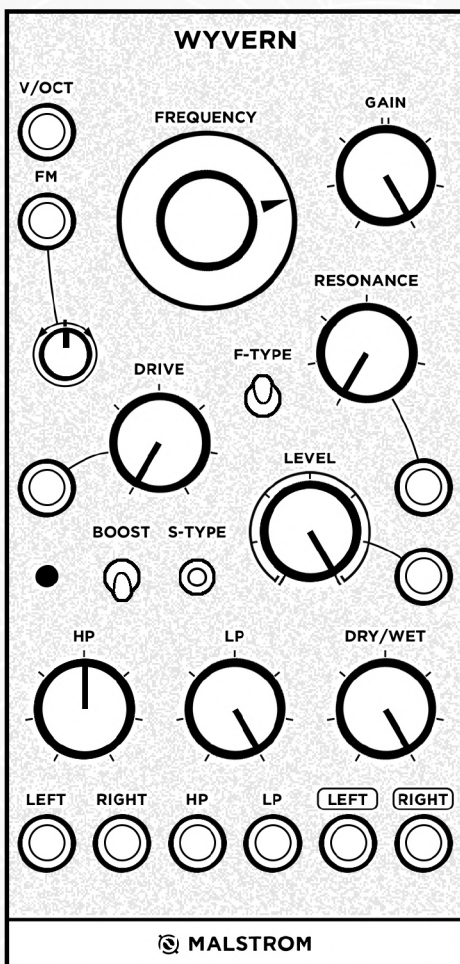
To create a Fuzz we want to generate lots of harmonics. So we use the Resonance to add a new sine wave, which we can mix and distort with our input signal. The sinewave can be distorted by increasing the Gain Knob.

While the mix of the two can be distorted with the Drive Knob. We use the Hard Clipping setting to get that "Fuzz" sound.

ADVANCED EQ

The following advanced EQ examples show how to expand on the EQ functionality of Wyvern, without using the Saturation (see note). It can generate quite complex EQ curves to help you get a tight mix! To do so we will make use of the Filter EQ stage and the Post-Saturation stage.

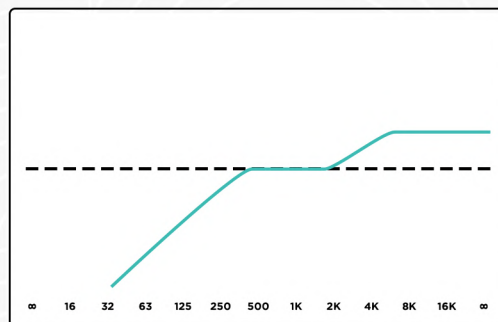
Note that the following graphs are exaggerated for readability.



HIGHPASS + HIGH SHELF

This combo can be used when a sound is muddy and lacking presence, could be processed recorded vocals for example.

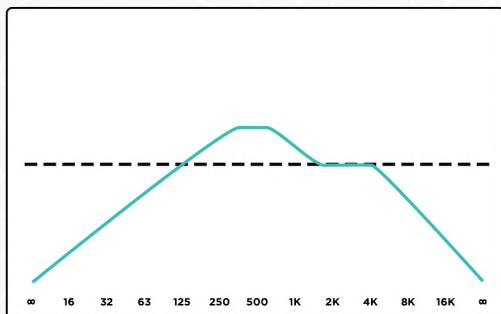
The lowpass removes the muddiness, while the high shelf adds presence and brightness.



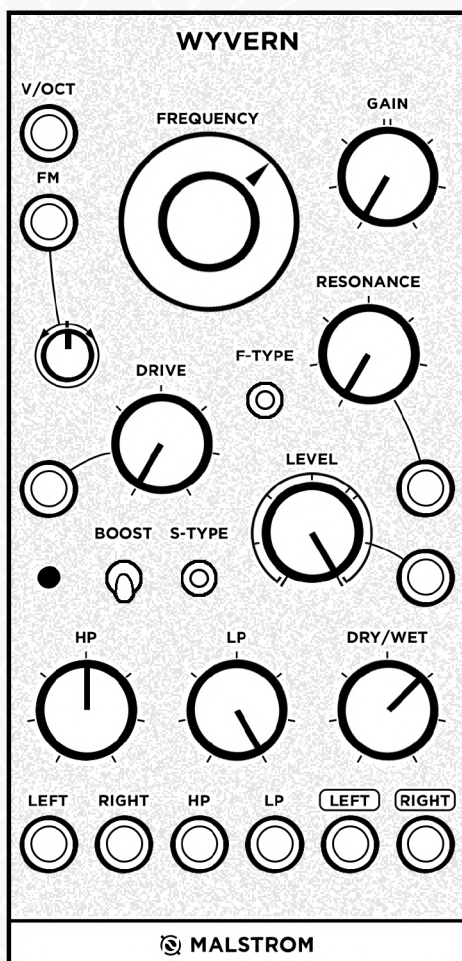
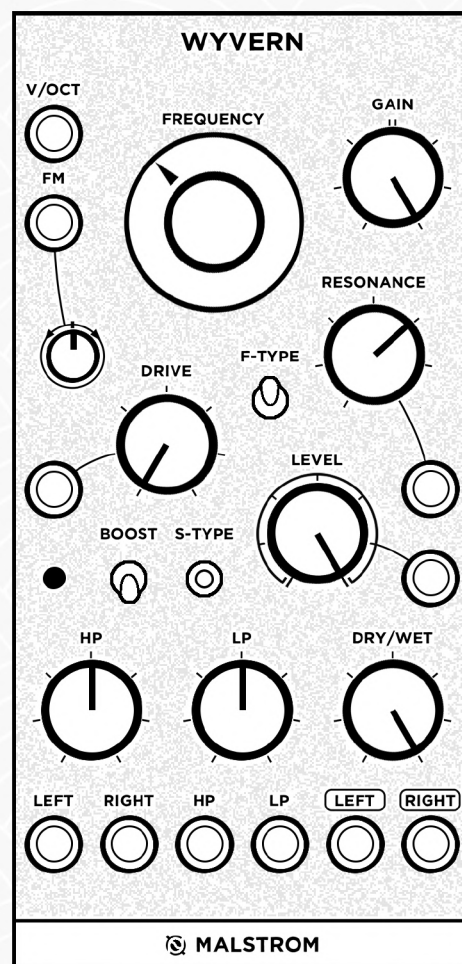
Note: Although we certainly recommend to try them with Saturation!

WIDE BANDPASS + LOW-MID BOOST

This combo can be used when you want a sound to sit in the mid-range of the spectrum, while also adding a bit of warmth.

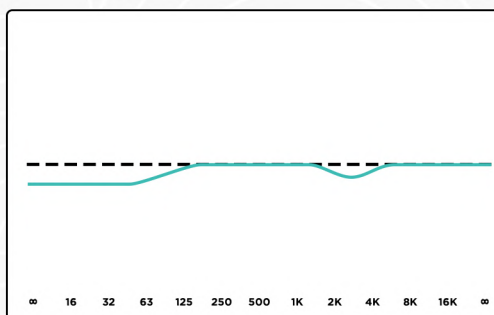


Tip: Resonance can help set the boost bandwidth. Alternatively switch to high shelf to boost high-mids instead!



LOW SHELF + NOTCH

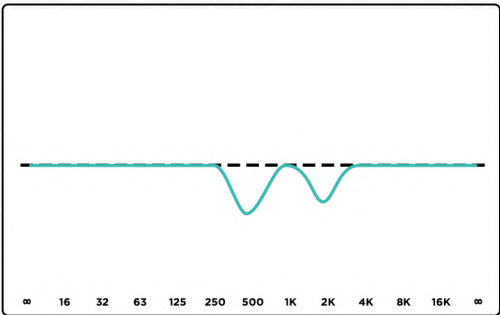
You do not need to go overboard with EQ, sometimes a little correction is all you need. In this case we will make use of the DRY/WET Knob to dial in the amount we want to remove. And turn that HP from Post into a low shelf.



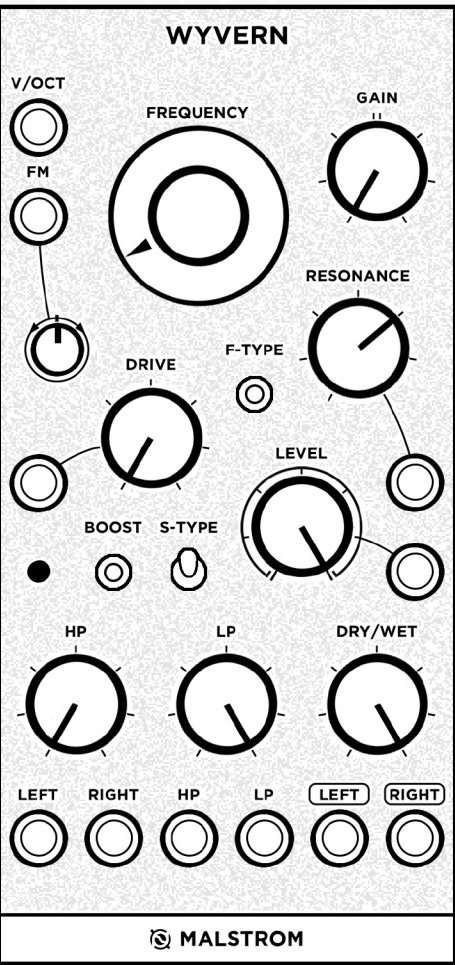
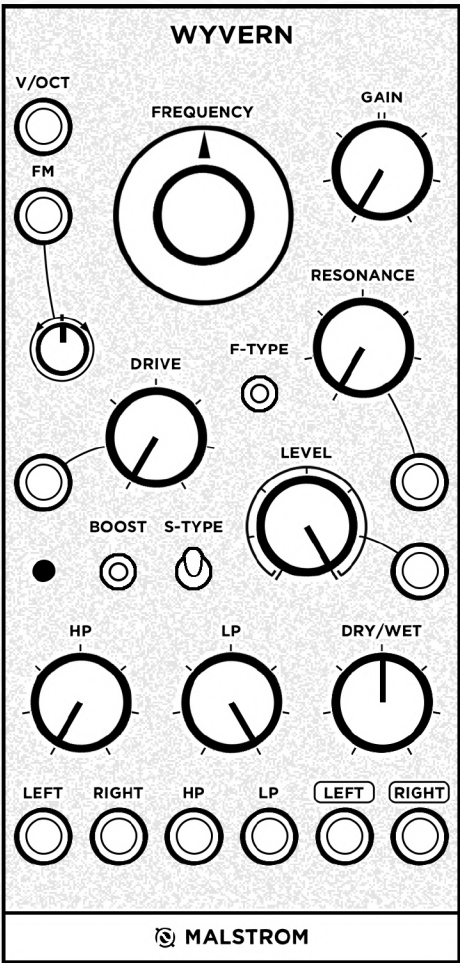
Tip: Keep your Gain Knob fully CCW or CW, this makes it much easier to set the amount with the DRY/WET Knob!

DOUBLE NOTCH

This EQ uses the Allpass distortion to add a phase shift, resulting in a double Notch. Use the Boost Switch to change the intensity and the Dry/Wet Knob to morph between single and double Notch.

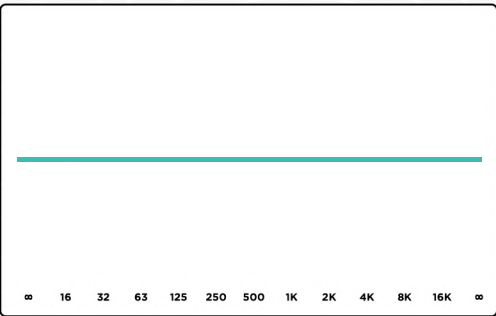


Tip: Set Resonance to full and set the Dry/Wet to the second line indicator to get a Resonating Double Notch!



PHASE ALTERATION

This uses the Resonance to cancel out the Notch, altering phase around the cutoff frequency while not changing the spectrum. Changing the Frequency will adjust the tone.

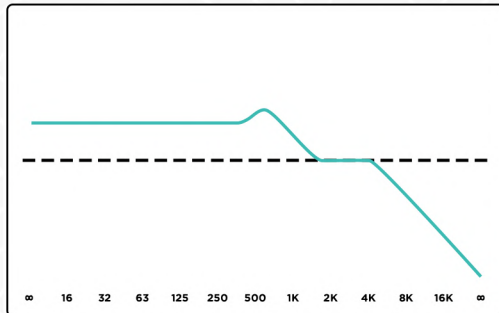


Tip: Setting the Frequency low and applying it on a bass voice is great!

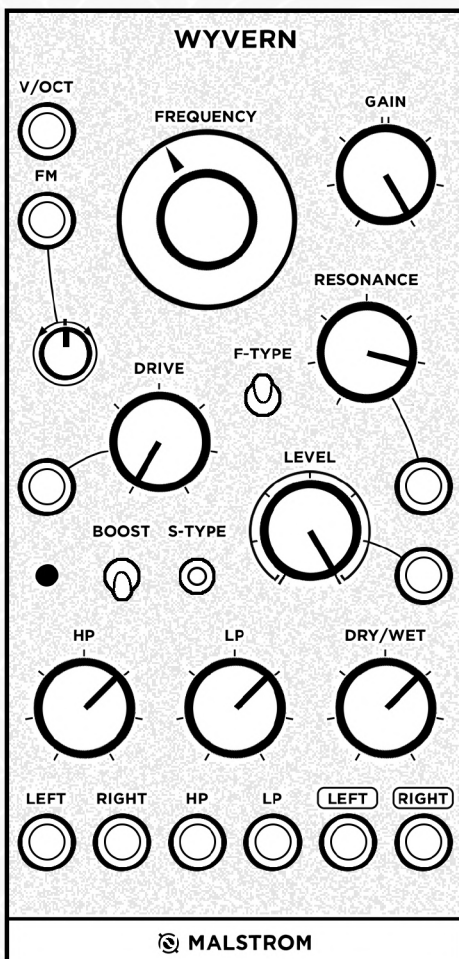
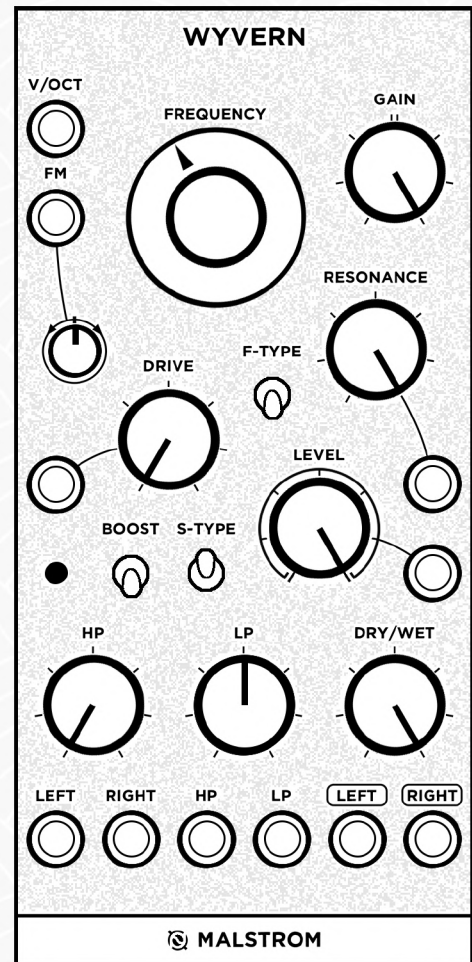
Tip: Change flavor with the S-Type switch!

PEAK + LOW SHELF + LOWPASS

We can use the Resonance to add a small peak at a specific frequency. Which works especially well when you use the Saturation stage too!

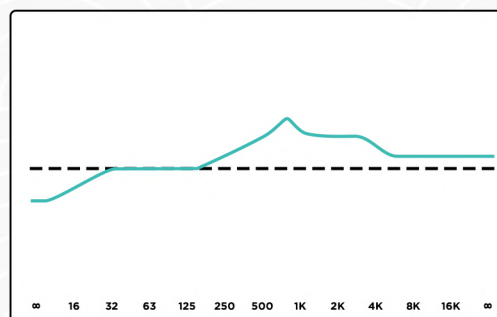


Tip: Check out page 19 for another way to use this creatively!



COMPLEX EQ

This is where we use all of the standard EQ options to get as complex as we can. This case could be a rumbly and distant mix that needs warmth and presence.



Tip: if you need to change a lot on your final mix, then it is always better to look at the source material!

PATCH EXAMPLES

EXAMPLE 1: KEY-TRACKING EQ

The following example shows a way to have your EQ follow the pitch of a synthvoice (or anything else that you are using with v/oct). A use case could be that you want to emphasize the higher harmonics of a sound. But as the sound goes higher in pitch; the EQ remains on the same frequency. Therefore lower harmonics are now also emphasized. By using this patch the EQ will maintain its frequency relation as the synthvoice changes in pitch.

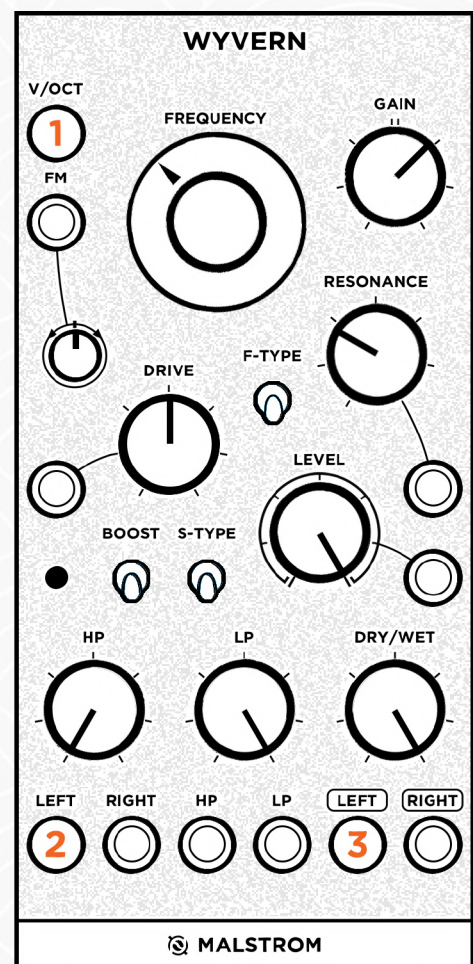
Start off by copying the knob and switch positions as shown in the example below. You will also need a buffered mult to patch!

PATCH

- 1a. Patch a v/oct signal into the V/Oct Input of your synthvoice using a mult.
- 1b. Patch the same v/oct signal into the V/Oct Input of Wyvern using a mult.
2. Patch the synthvoice output to the Wyvern Left Audio Input
3. Patch the Left Audio Output so you can listen.

NOTES

Set a Frequency that you like, and Increase or decrease the Gain Knob.



Tip: Continue by adding Resonance and Saturation!

EXAMPLE 2: SHIFTED RESONANCE FILTER

This patch shows how you can create a filter where the slope frequency and Resonance frequency are different from each other. Emphasizing a specific frequency but not cutting straight after it. In the example we will be making a lowpass filter.

Start off by copying the knob and switch positions as shown in the example below. A buffered mult may be handy if you are looking for precision.

PATCH

1. Patch an audio source into the Left Audio Input.
2. Patch a modulation source into the V/Oct Input.
3. Patch the same modulation source into the LP CV Input.
4. Patch the Output to listen.

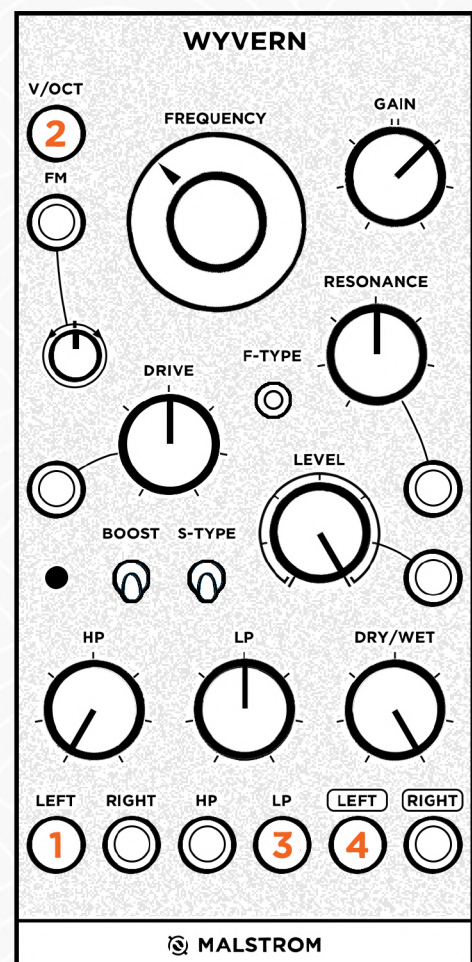
NOTES

The Frequency determines the frequency of your peak. The peak amount can be set with the Gain Knob and/or Resonance.

The LP Knob determines the cutoff frequency of your filter.

Now as the V/Oct Input and LP CV Input are modulated they will keep the frequency ratio you have set!

Tip: Use a macro controller if you want to control the filter by hand!



EXAMPLE 3: PHASE MODULATION

This patch shows how Wyvern can be patched to create a Phaser & Distortion effect. A Phaser effect is made from an Allpass Filter of which the time is modulated. The result is then mixed with the original signal.

The most important setting to use is the Saturation Type: Allpass Distortion. Start off by copying the knob and switch positions as shown in the example below. You will also need an lfo!

PATCH

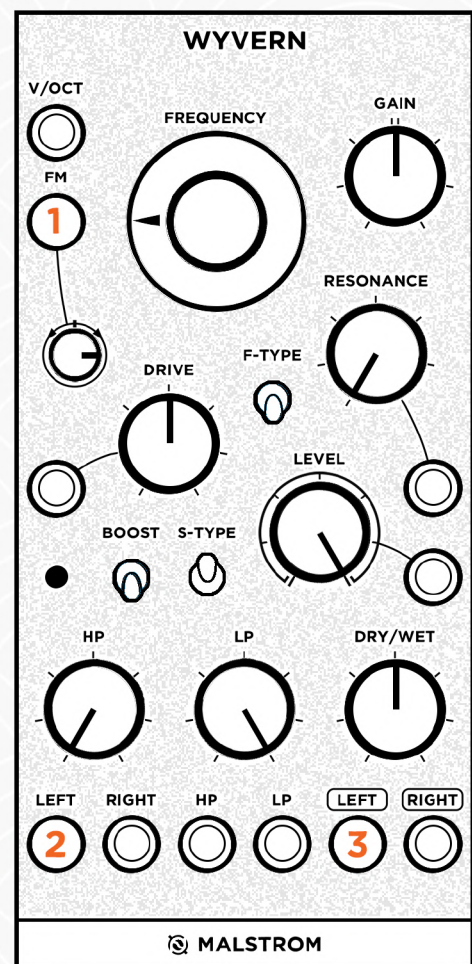
1. Patch an lfo into the FM CV Input.
2. Patch an audio source into the Left Audio Input.
3. Patch the Left Audio Output to listen.

NOTES

The Frequency Knob sets the Allpass Filter's time delay.

Play around with the Drive Knob to increase Distortion amount. And adjust the Dry/Wet to your liking.

Tip: Very much recommended on bass synths. Set F-Type to mid position and turn the Gain Knob fully CCW for notch filter. Turn Frequency Knob fully CCW, now turn it up slowly and search for your sweet spot!



V/OCT CALIBRATION

The following instructions are needed to calibrate the v/oct scaling. However Wyvern comes calibrated, so only do this if you have the right tools and consider it necessary!

To calibrate you will need a calibrated voltage source, a digital tuner and a flathead screwdriver.

Locate the trimmers on the pcb indicated with V/OCT. Start with the one marked as LEFT. Make sure you have access to it. Turn the power on and let the module warm up for approximately 20 minutes.

Go ahead and copy the knob positions in the example on the right.

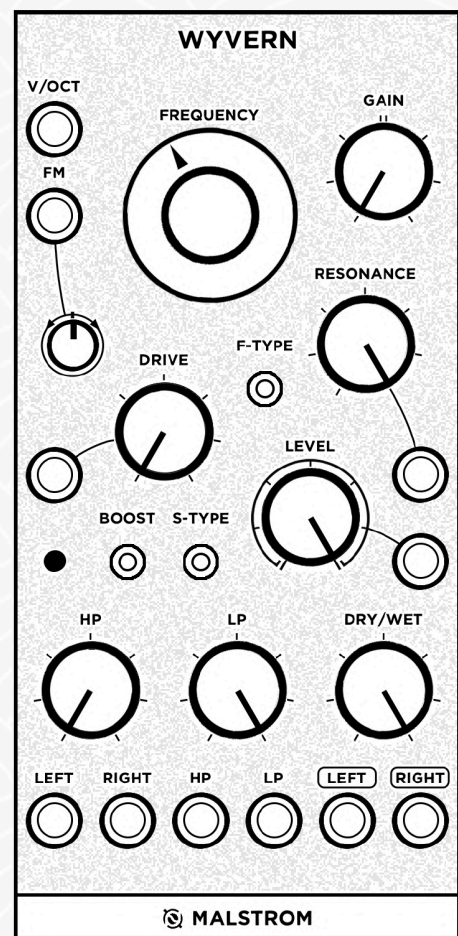
1. Patch the v/oct source in the V/Oct Input.
2. Patch the Output to your digital tuner.

Set the v/oct source to D0 and tune the Frequency knob to A0.

Going back and forth between D0 and D2. Wyvern should give A0 and A2 as output. Make sure that there is perfect tuning between them by adjusting the pcb trimmer.

If the A0 output is flat: the pcb trimmer needs to go CCW. If the A0 is sharp: the pcb trimmer needs to go CW. Tune Frequency knob to A0 again and repeat these instructions.

Note that you will need to copy the instructions for the right channel. Do not worry if the channels are not in tune, this step is shown on page 22.



TUNE CHANNELS LEFT AND RIGHT

It is best to tune both channels by ear. Copy the knob positions from the previous page. Listen to the left channel on your left ear and right channel on your right ear. Adjust the TUNE trimmer and make sure there is as little “beating” between them as possible.

SET RESONANCE AMOUNT

You can set or limit the amount of resonance by adjusting the RES-LIM trimmer on the pcb. To adjust it copy the knob positions on the previous page first. Standard setting is right after point of oscilation (+-5V).

ADJUST GAIN KNOB CENTER POSITION

You can adjust the point where the gain knob has its center position. Goal is to get as little bleed in the center when moving from right to the center and from left to the center. You will need to do this for both the left and right channel. You can do so by copying the knob positions on the previous page first. And adjust the CTR-ADJ trimmers. The one next to the CTR-ADJ Left mark is for the left channel.

I/O SPECS

- All CV inputs DC coupled.
- All inputs handle audio signals.
- All inputs have 100k input impedance.
- All inputs have +-10V Input range.
- All inputs are overvoltage protected.
- Temperature compensated V/Oct Input.
- Volume CV input expects 0-5V voltage range.
- Keyed power header and reverse power protection.

WARRANTY & SUPPORT

WARRANTY

We guarantee that all of our products are tested before shipping. However it may be that you are experiencing problems with a product. Do not worry we are here to help! Within one year after purchase, any damaged or malfunctioning products will be repaired, serviced and calibrated on a return-to-factory basis. However this service is only guaranteed if the damages are not caused during shipping, installation, incorrect usage or user inflicted damages.

Even after the one year warranty, or if the damages are caused by the before mentioned causes; we want to help you. Products will still be repaired but we reserve the right to charge you for labor, parts and transit expenses.

In any case we advise you to contact us about any issues!

SUPPORT

For any support head over to our support page:
malstromaudio.com/support.

Or send us a mail at info@malstromaudio.com and we will get back to you as soon as possible.

SPECIAL THANKS

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